

Figure 1

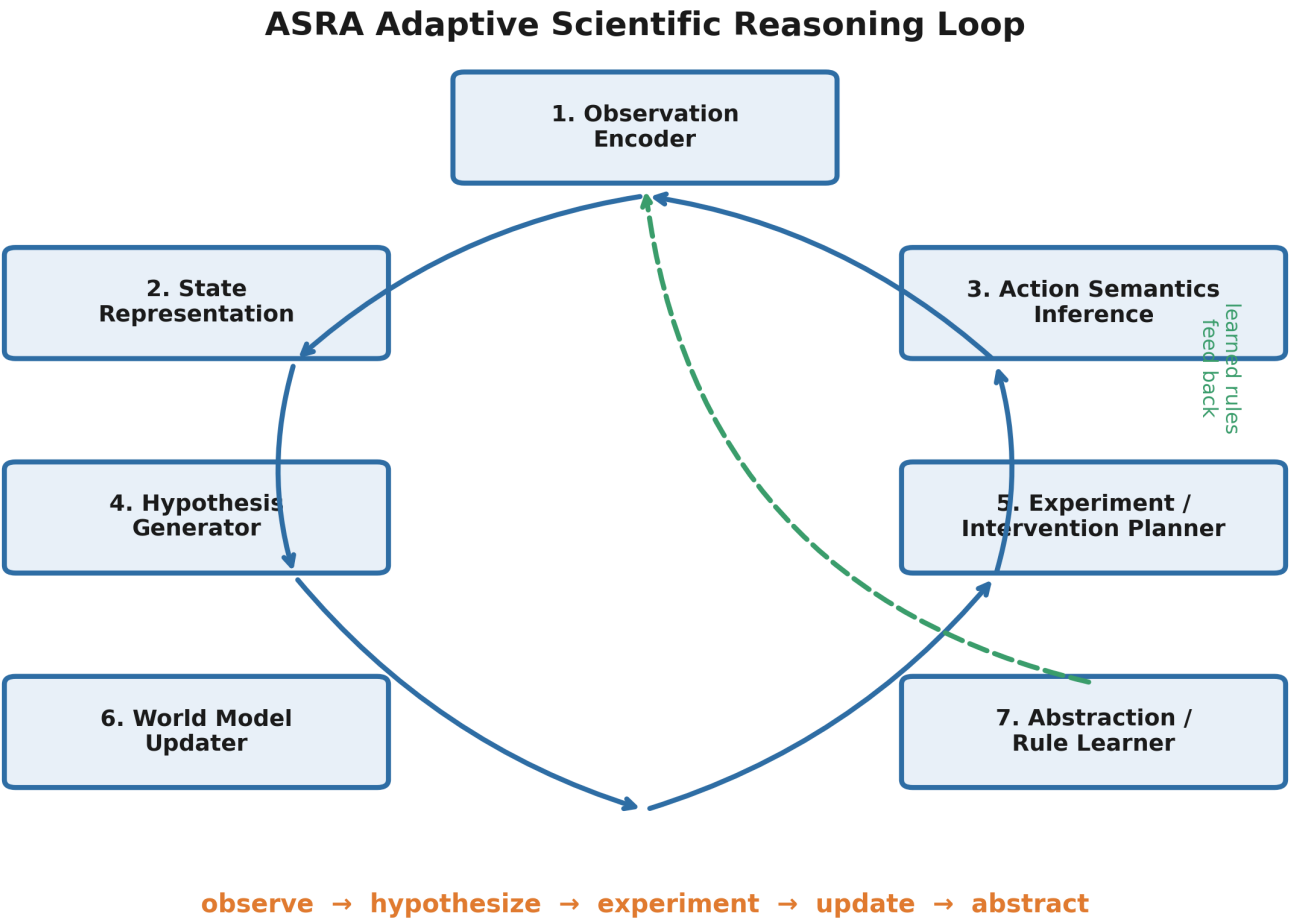


Figure 1. ASRA architecture overview. The system observes state transitions, infers action semantics, generates hypotheses, selects experiments, updates a world model, and abstracts reusable rules through a recurrent reasoning loop.

Figure 2

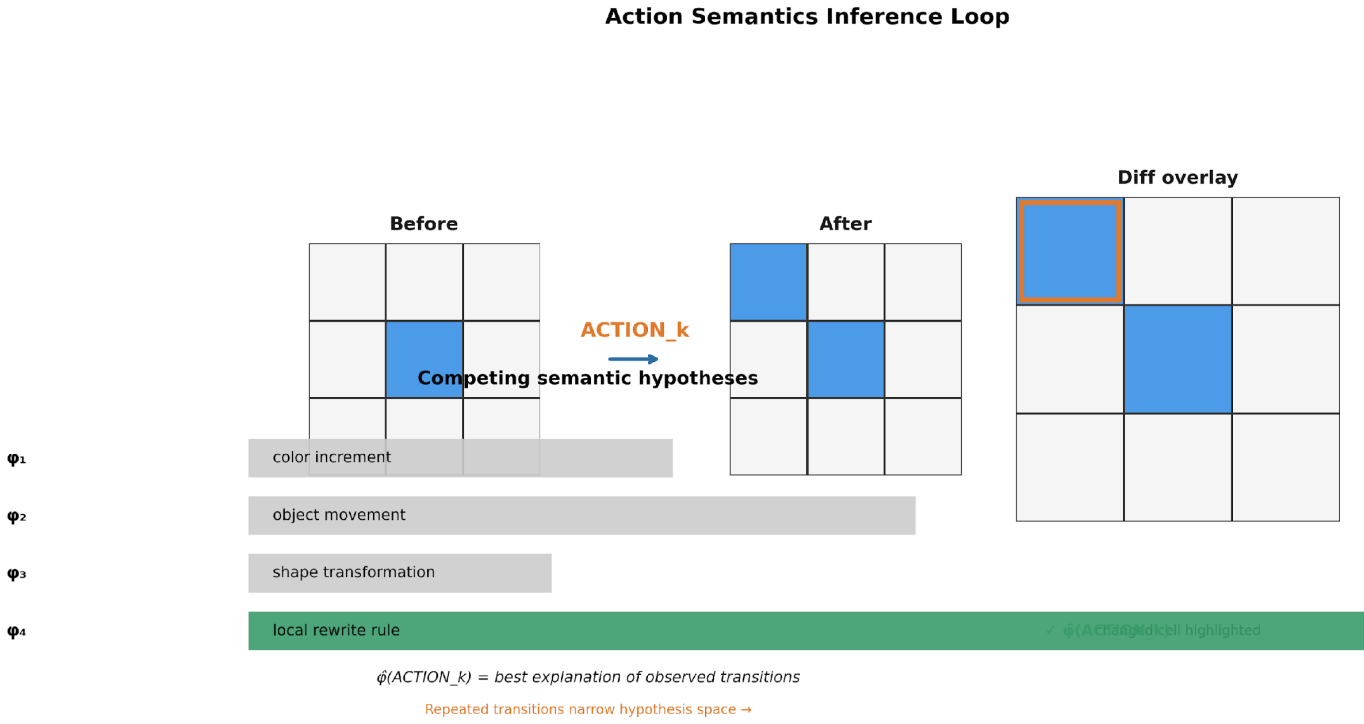


Figure 2. Action semantics inference loop. ASRA observes before-and-after state transitions, computes differences, evaluates competing semantic hypotheses, and converges on the inferred meaning of an unknown action token.

Figure 3

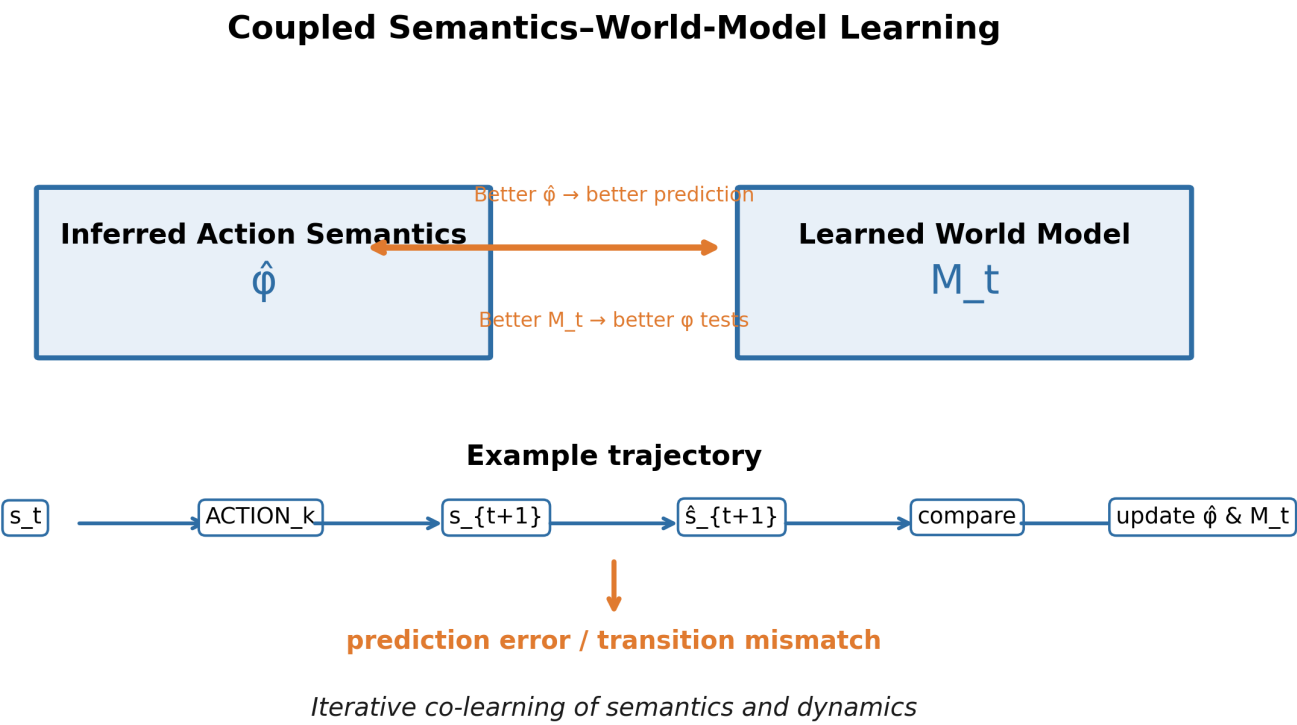


Figure 3. Coupled semantics-world-model learning. ASRA jointly refines action meanings and transition models by comparing predicted and observed next states across repeated trajectories.

Figure 4

Decision Biology Mapping

ASRA Concept	Grid / Prototype	Decision Biology	Dataset / Measurement
State	Grid configuration	Cellular state	Transcriptomics, proteomics, ima
Action	ACTION_k	Perturbation	Drug, CRISPR, ligand, stressor
Transition	Before → after grid	Pre/post response	Time-series omics
Action semantics	Inferred grid rule	Perturbation mechanism	Pathway response, gene regulat
World model	Transition predictor	Biological response model	LINCS, scPerturb, Perturb-seq
Experiment planner	Next grid test	Next intervention	Active learning / experimental d
Abstraction	Reusable transform rule	Biological decision principle	Adaptation, resistance, different

Figure 4. Decision Biology mapping. ASRA's state-transition reasoning framework maps naturally onto biological perturbation-response systems, where actions correspond to interventions and state changes correspond to cellular responses.

Figure 5

ASRA Prototype Environment

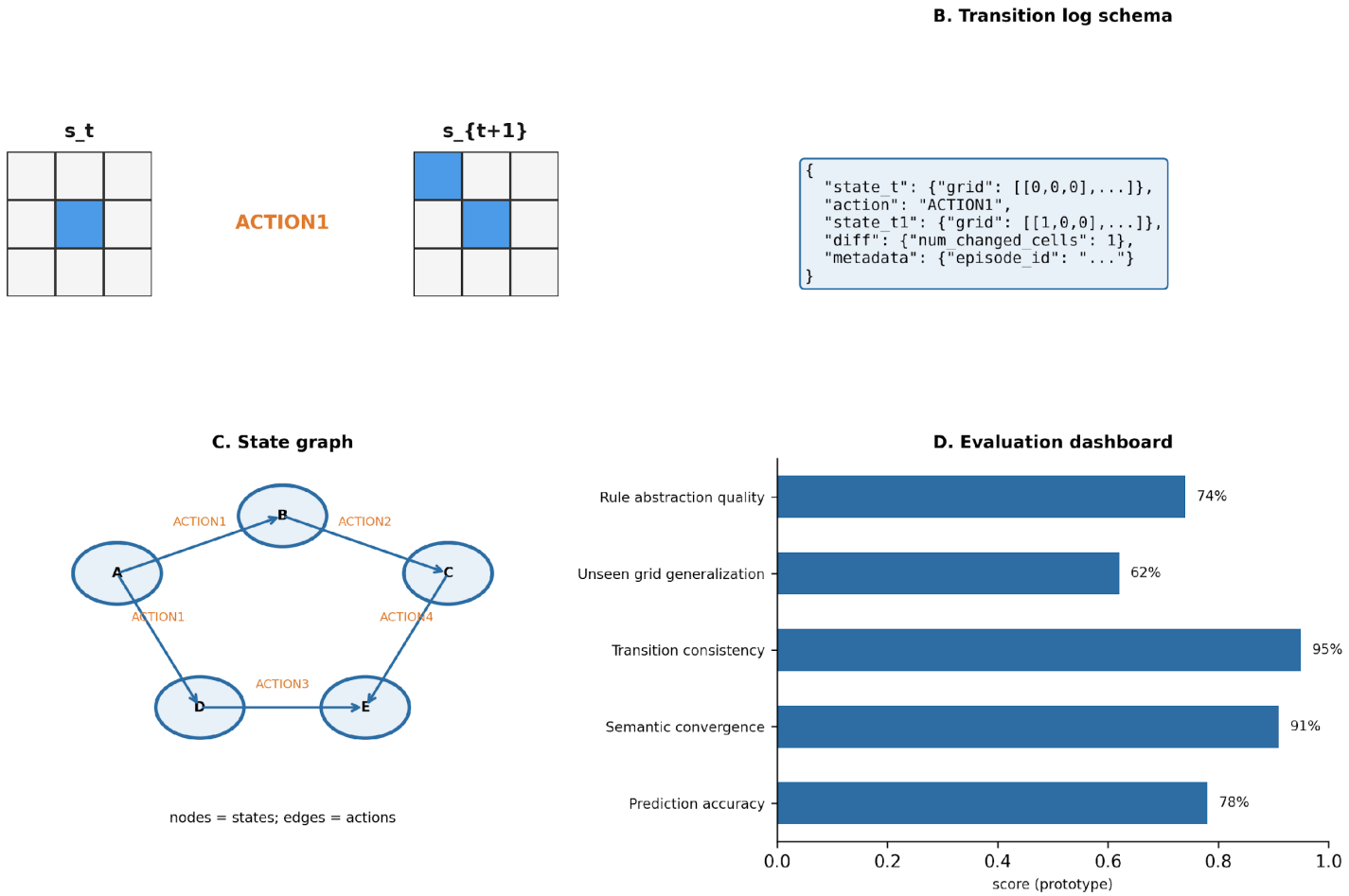


Figure 5. Prototype environment. The ASRA prototype uses grid-world transitions, structured transition logs, state graphs, and evaluation dashboards to test action semantics inference and adaptive world-model learning.